

Project 3

Designing Your Defensive Monitoring Environment

Project Week



Welcome

This Week:

In this project, you will play the role of an SOC analyst at a fictional organization called Virtual Space Industries (VSI).

01

VSI is a company that specializes in the design of virtual-reality programs for businesses.

02

VSI has heard rumors that a competitor, JobeCorp, may launch cyberattacks to disrupt VSI's business.

03

As an SOC analyst, you are tasked with using Splunk to monitor against potential attacks on your systems and applications.



For your third project,
you will use the skills you
learned in the Defensive
Security unit to design a
custom monitoring
environment to protect a
fictional organization.





Defensive Security

In the Defensive Security unit, we've covered:

- 1 Information security continuous monitoring (ISCM)
- 2 Log types and how they are used for monitoring
- 3 Log aggregation and correlation
- 4 Baselineing
- 5 SIEMS
- 6 Splunk

Group Work

This is a group project.

- You will complete this project in groups, but every student is required to remain in class.
- Groups are permitted to split up the work on this project, but each student must submit all project deliverables.
- Each day builds on the previous day's work to complete the project, and each day's activities must be completed in order.



This Week:

You are tasked with monitoring the following VSI products:

01

An administrative webpage:

<https://vsi-corporation.azurewebsites.net/>

02

An Apache web server,
which hosts this webpage.



03

A Windows operating system,
which runs many of VSI's back-end operations.



Your networking team has also provided you with past logs to help you develop baselines and create reports, alerts, and dashboards to protect VSI from any attacks by JobeCorp.

This Week's Daily Structure

Class will run a little differently during project week. Each day will include:

Brief Lecture

We will begin by introducing some new concepts.

Brief Overview

Then, we'll review the day's project tasks.

Daily Guides

Finally, you'll use a guide to complete the project tasks during the remaining class time.

Day 2's guide concludes with review questions, which you'll submit at the end of the project.



Daily Objectives and Milestones

Day 1

You will develop a defensive solution utilizing a variety of Splunk tools you learned in class in order to protect VSI.

1

You will be given logs of “typical” business functions in order to understand VSI's environment.

2

You will use these logs to create baselines and then design custom alerts, reports, and dashboards.

Additionally, you will download and use a Splunk “add-on” app of your choice to monitor against other types of attacks.



Daily Objectives and Milestones

Day 2

VSI will
experience
a simulated
cyber attack.

1

You will analyze the reports and dashboards that you created on Day 1 in order to determine whether your defensive choices protected VSI from these attacks.

2

You will be provided with an answer review and analysis questions.

Additionally, you will start preparing slides to present your findings on Day 3 of class.



Daily Objectives and Milestones

In your groups, you will present:

Day 3

You will present the defensive solution you created.

1

You will discuss how well or poorly it alerted and protected against the simulated attacks.

2

You will suggest any adjustments you would make to your defensive solution.

Project Deliverables

The project is due one week from the third day of this project. There is no additional Challenge this week. You will submit the following project deliverables:

Technical Brief and Review Questions

Provide answers to a series of questions explaining your monitoring solution and its efficacy at defending against attacks. Screenshots of your monitoring solution will also be submitted with this document.

Presentation Slides

Submit the slides your group uses for Day 3's presentation.

Today's Class

The rest of today's class will proceed as follows:

01

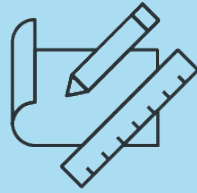
Introduction to the
project scenario

02

Overview of
today's tasks

03

Project work



Day 1 **Overview**

Day 1 Project Overview

On Day 1, you will complete the following:

01

Load and analyze Windows logs.

02

Create reports, alerts, and dashboards for the Windows logs.

03

Load and analyze Apache logs.

04

Create reports, alerts, and dashboards for the Apache logs.

05

Install an add-on Splunk application for additional monitoring.



Step 1

Load and Analyze Windows Logs

You will be provided with a set of logs from the Windows servers that run VSI's back-end systems.

The logs represent normal business operations.

You will load these logs and analyze the data they contain to determine baselines of normal activity.

The screenshot displays the Splunk Enterprise web interface. At the top, the navigation bar includes 'Search', 'Analytics', 'Datasets', 'Reports', 'Alerts', and 'Dashboards'. The 'Search' tab is active, showing a 'New Search' page. The search query is `source="windows_server_logs.csv" host="0dcce57faf30" sourcetype="csv"`. Below the query, it indicates '4,764 events' and 'No Event Sampling'. The 'Events (4,764)' tab is selected, showing a timeline visualization with green bars. Below the timeline, a table of events is displayed. The table has columns for 'Time' and 'Event'. The first event is dated '3/24/20' at '11:59:54.000 PM'. The event details show a security event from 'Domain_A' where a user account was deleted. The interface also shows 'SELECTED FIELDS' (host, source, sourcetype) and 'INTERESTING FIELDS' (Account_Domain, Account_Name).

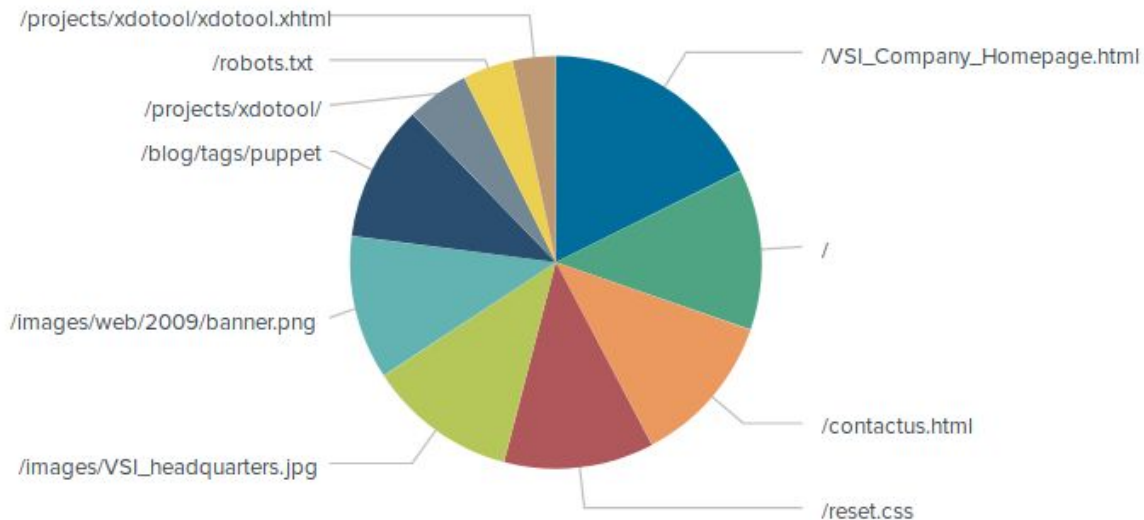
i	Time	Event
>	3/24/20 11:59:54.000 PM	2020-03-24T23:59:54.000+0000,,"Domain_A Domain_A",,"user_f user_l",,,,,,,,,Account Management,,,,,,,,ACME-002,,,,,,,,-4726,A user account was er account was deleted. Subject: Security ID: Domain_A\user_f Show all 63 lines host = 0dcce57faf30 source = windows_server_logs.csv sourcetype = csv



Step 2

Create Reports, Alerts, and Dashboards for the Windows Logs

Then, you will create reports, alerts, and dashboards based on your research in the previous step. You'll use these reports, alerts, and dashboards to determine if a future attack occurs.



Important:

You must grab screenshots where indicated in the daily guide. You will add them to your presentation.

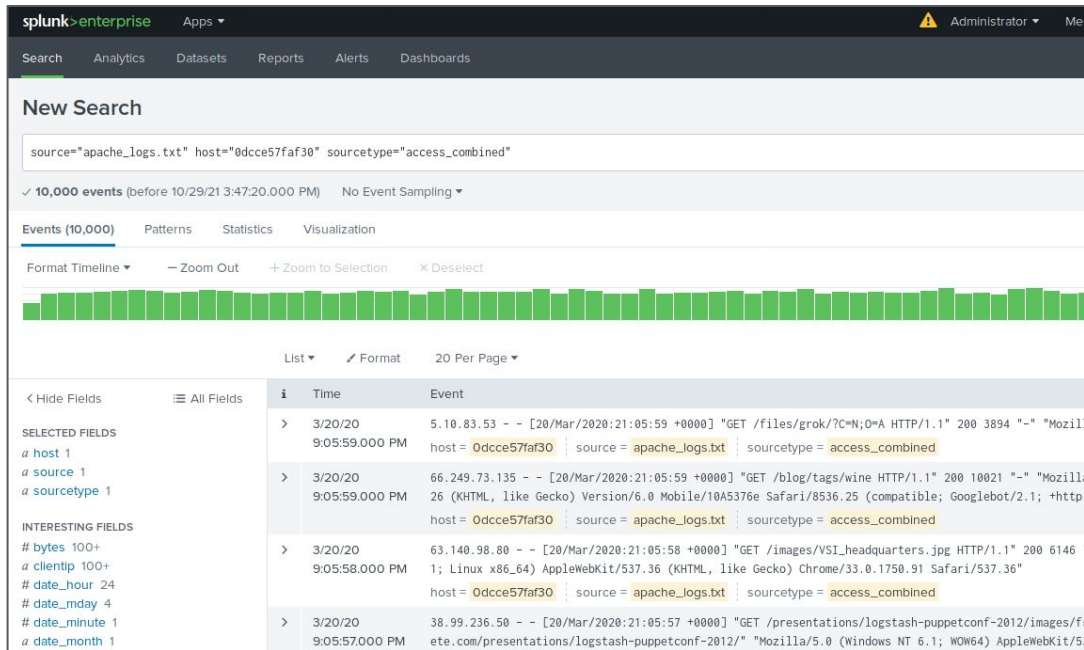


Step 3

Load and Analyze Apache Logs

Next, you will be provided with a set of logs from the Apache web servers that host VSI's web application. The logs represent normal business operations.

You will load these logs and analyze the data they contain to determine baselines of normal activity.

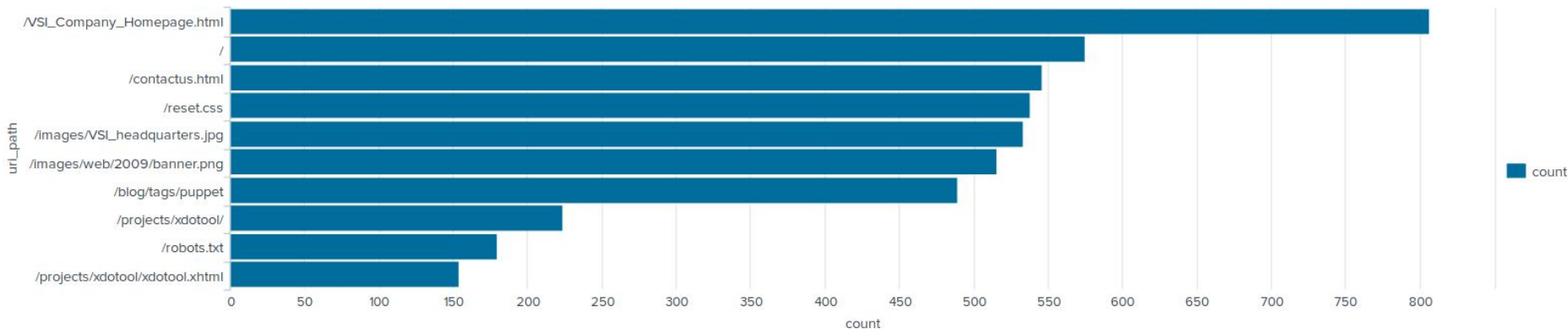




Step 4

Create Reports, Alerts, and Dashboards for the Apache Logs

Then, you will create reports, alerts, and dashboards based on your research in the previous step. You'll use these reports, alerts, and dashboards to determine if a future attack occurs against VSI's web server.





Step 5

Install Add-On Splunk Applications for Additional Monitoring

You will choose one of several Splunk add-on apps to assist with monitoring. You will install the app and configure it to protect VSI's systems.

**Splunk Add-on for Unix and Linux**
By Splunk Inc.

*** Important: Read upgrade Instructions and test add-on update before deploying to production *** There are changes to...

PLATFORM Splunk Enterprise, Splunk Cloud

RATING ★★★★★ 4 (40)

 **SPLUNK SUPPORTED ADDON**

**Splunk Common Information Model (CIM)**
By Splunk Inc.

The Common Information Model is a set of field names and tags which are expected to define the least common...

PLATFORM Splunk Enterprise, Splunk Cloud

RATING ★★★★★ 5 (19)

 **SPLUNK SUPPORTED ADDON**

**Palo Alto Networks Add-on for Splunk**
By Palo Alto Networks

The Palo Alto Networks Add-on for Splunk allows a Splunk® Enterprise administrator to collect data from every...

PLATFORM Splunk Enterprise, Splunk Cloud

RATING ★★★★★ 4 (19)

 **DEVELOPER SUPPORTED ADDON**



Group Assignments



Activity:

Building a Security Monitoring Environment — Day 1

In this project, you will design reports, alerts, and dashboards, and select and install a Splunk add-on app in order to protect VSI from attacks.

Suggested Time:
To end of class





Questions?





Thank you